PROJECT REPORT

Car Rental Request Automation

1. INTRODUCTION

1.1 Project Overview

Car Rental Request Automation is a ServiceNow-based project designed to automate the process of submitting, processing, approving, and tracking car rental requests. The project uses Service Catalog and Workflow to replace manual request handling with a structured and automated process.

1.2 Purpose

The main purpose is to provide employees with a simple way to request a rental car through the ServiceNow Service Catalog. The workflow automatically routes the request, manages approval, creates the required tasks, sends notifications, and updates the request status.

2. IDEATION PHASE

2.1 Problem Statement

Manual car rental requests can involve emails, phone calls, repeated data entry, delayed approvals, and difficulty tracking request status. There is a need for a centralized system that standardizes the request process and reduces manual work.

2.2 Empathy Map Canvas

- User needs a simple form to submit car rental requirements.
- User wants to know whether the request is approved and its current status.
- Approver needs complete request information to make a decision.
- Support or rental team needs clear tasks and request details.

2.3 Brainstorming

- Create a Car Rental Request catalog item.
- Capture employee, rental date, location, vehicle type, and purpose.
- Automate approval through ServiceNow Workflow.
- Create fulfillment tasks after approval.
- Send notifications for submission, approval, rejection, and completion.
- Allow request tracking from ServiceNow.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Request Creation → Request Submission → Manager/Authorized Approval → Fulfillment Task → Rental Processing → Request Completion. If the request is rejected, the workflow updates the request accordingly and notifies the requester.

3.2 Solution Requirement

- ServiceNow instance with Service Catalog and Workflow capabilities.
- Catalog item for submitting a car rental request.
- Variables for required rental information.
- Workflow for approval and fulfillment.
- Notifications for important request status changes.
- Request and task records for tracking and audit purposes.

3.3 Data Flow Diagram

Requester → Car Rental Catalog Item → ServiceNow Request/RITM → Workflow → Approval → Fulfillment Task → Rental Team → Completion/Notification.

3.4 Technology Stack

- Platform: ServiceNow
- Service Catalog: Used to create and submit the car rental request.
- Workflow: Used to automate approval, task creation, notifications, and status movement.
- ServiceNow Tables: Request/Requested Item and task-related records.
- ServiceNow Scripting/Configuration: Used where required for validation and automation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution centralizes the complete car rental request process in ServiceNow. It reduces dependency on manual communication, provides consistent request information, improves visibility, and makes the approval and fulfillment process easier to track.

4.2 Proposed Solution

A Car Rental Request catalog item is created in ServiceNow. The requester enters the required details and submits the request. A workflow receives the request, sends it for approval, and based on the decision either stops the request with a rejection notification or continues to fulfillment. After approval, the required task is assigned to the responsible team. Once the task is completed, the request is closed and the requester is notified.

4.3 Solution Architecture

- Requester Layer: Employee accesses the Service Catalog and submits the car rental request.
- Catalog Layer: Car Rental Request item collects and validates required information.
- Workflow Layer: Controls approval, decision, task creation, notifications, and completion.
- Approval Layer: Authorized approver reviews and approves/rejects the request.
- Fulfillment Layer: Rental/support team receives and completes the generated task.
- Tracking Layer: ServiceNow request records maintain status and history.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Phase 1: Identify the manual car rental request process and required information.

- Phase 2: Design the Service Catalog item and request variables.
- Phase 3: Configure the workflow and approval process.
- Phase 4: Configure task assignment and notifications.
- Phase 5: Test approval, rejection, fulfillment, and completion scenarios.
- Phase 6: Review the complete automated request flow and document the project.

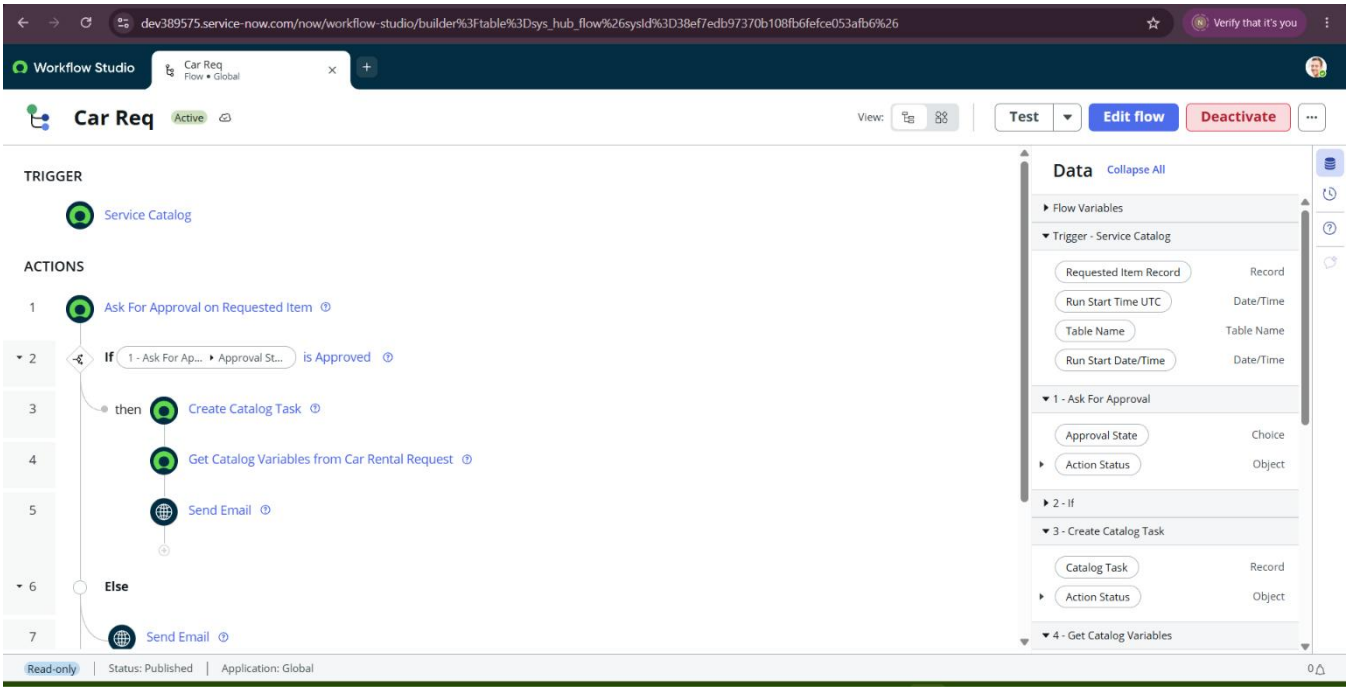
6. FUNCTIONAL AND PERFORMANCE TESTING

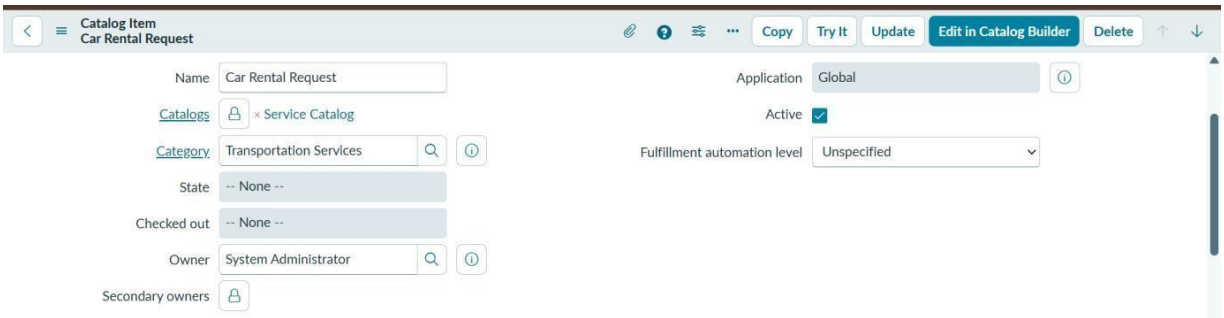
6.1 Performance Testing

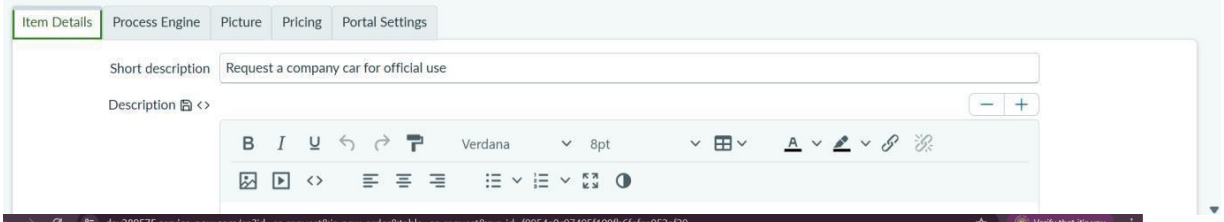
Functional testing focuses on whether the catalog item accepts valid request information, required fields are enforced, the workflow starts correctly, approval decisions are handled correctly, fulfillment tasks are created after approval, and notifications/status updates occur as expected. Performance is evaluated from the perspective of smooth form submission and timely workflow processing within the ServiceNow environment.

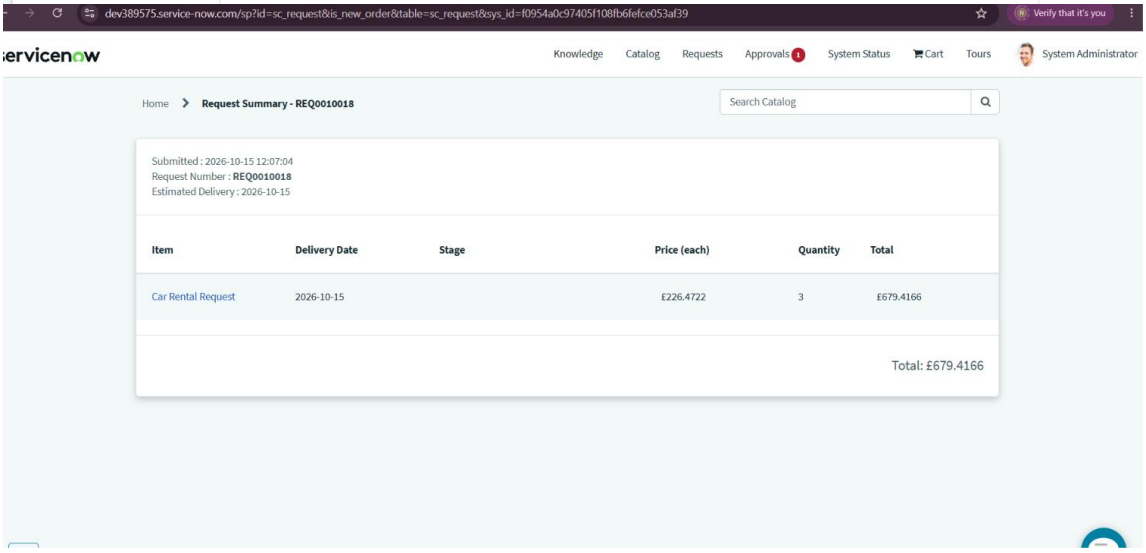
- Test valid car rental request submission.
- Test missing mandatory information.
- Test approval path.
- Test rejection path.
- Test fulfillment task creation.
- Test request completion and status update.
- Test requester notifications.

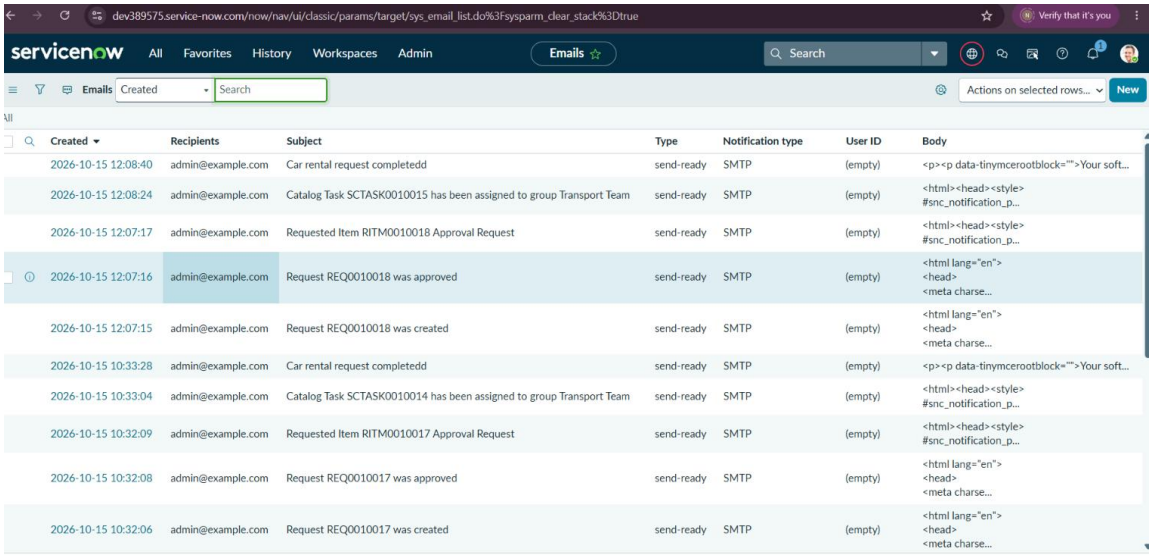
7. RESULTS

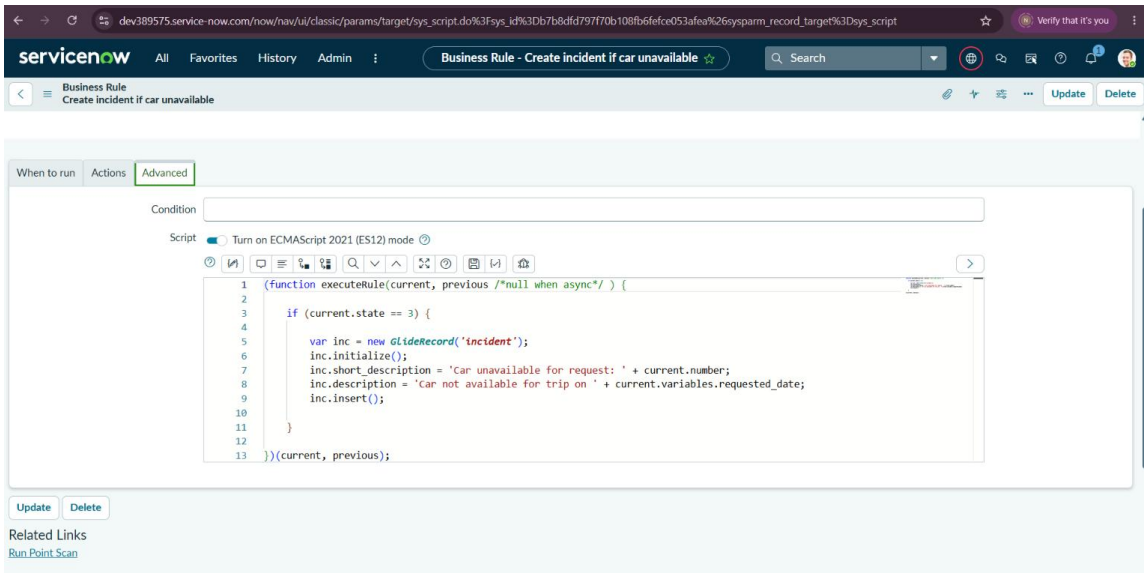



Catalog Item
Car Rental Request
Copy
Try It
Update
Edit in Catalog Builder
Delete
Name
Car Rental Request
Application
Global
Catalogs
Service Catalog
Active
Category
Transportation Services
Fulfillment automation level
Unspecified
State
-- None --
Checked out
-- None --
Owner
System Administrator
Secondary owners


Item Details
Process Engine
Picture
Pricing
Portal Settings
Short description
Request a company car for official use
Description
Verdana
8pt


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servicenow
Knowledge
Catalog
Requests
Approvals
System Status
Cart
Tours
System Administrator
Home
Request Summary - REQ0010018
Search Catalog
Submitted : 2026-10-15 12:07:04
Request Number : REQ0010018
Estimated Delivery : 2026-10-15
Item
Delivery Date
Stage
Price (each)
Quantity
Total
Car Rental Request
2026-10-15
£226.4722
3
£679.4166
Total: £679.4166


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servicenow
All
Favorites
History
Workspaces
Admin
Emails
Search
Emails
Created
Search
Actions on selected rows...
New
Created
Recipients
Subject
Type
Notification type
User ID
Body
2026-10-15 12:08:40 admin@example.com Car rental request completedd send-ready SMTP (empty)
2026-10-15 12:08:24 admin@example.com Catalog Task SCTASK0010015 has been assigned to group Transport Team send-ready SMTP (empty)
2026-10-15 12:07:17 admin@example.com Requested Item RITM0010018 Approval Request send-ready SMTP (empty)
2026-10-15 12:07:16 admin@example.com Request REQ0010018 was approved send-ready SMTP (empty)
2026-10-15 12:07:15 admin@example.com Request REQ0010018 was created send-ready SMTP (empty)
2026-10-15 10:33:28 admin@example.com Car rental request completedd send-ready SMTP (empty)
2026-10-15 10:33:04 admin@example.com Catalog Task SCTASK0010014 has been assigned to group Transport Team send-ready SMTP (empty)
2026-10-15 10:32:09 admin@example.com Requested Item RITM0010017 Approval Request send-ready SMTP (empty)
2026-10-15 10:32:08 admin@example.com Request REQ0010017 was approved send-ready SMTP (empty)
2026-10-15 10:32:06 admin@example.com Request REQ0010017 was created send-ready SMTP (empty)


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Verify that it's you
servicenow
All
Favorites
History
Admin
Business Rule - Create incident if car unavailable
Search
Business Rule
Create incident if car unavailable
Update
Delete
When to run
Actions
Advanced
Condition
Script
Turn on ECMAScript 2021 (ES12) mode
(function executeRule(current, previous /*null when async*/) {
if (current.state == 3) {
var inc = new GlideRecord('incident');
inc.initialize();
inc.short_description = 'Car unavailable for request: ' + current.number;
inc.description = 'Car not available for trip on ' + current.variables.requested_date;
inc.insert();
}
})(current, previous);
Update
Delete
Related Links
Run Point Scan

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work and repeated communication.
- Provides a centralized request and tracking process.
- Automates approval and fulfillment activities.
- Improves visibility of request status.
- Creates a consistent and auditable process.

Disadvantages

- Requires initial ServiceNow configuration and workflow setup.
- Changes to business rules may require workflow updates.
- The solution depends on the availability and configuration of the ServiceNow platform.

9. CONCLUSION

The Car Rental Request Automation project demonstrates how ServiceNow Service Catalog and Workflow can be used to automate a common business request process. The solution provides a structured flow from request submission to approval and fulfillment, reducing manual effort and improving request visibility and control.

10. FUTURE SCOPE

- Integrate with external car rental or fleet management systems.
- Add automatic vehicle availability checking.
- Add cost estimation and budget validation.
- Support multiple vehicle categories and rental policies.

- Add dashboards and reports for request volume, approval time, and fulfillment performance.
- Add automated reminders for pending approvals and overdue fulfillment tasks.

11. APPENDIX

Source Code: ServiceNow scripts are maintained in the configured Business Rule and portal widgets.

https://github.com/shreee8/Car-Rental-Request-Automation

Servicenow Instance: https://dev389575.service-now.com/sp